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HARMONISED EUROPEAN STANDARD

Satellite Earth Stations and Systems (SES);
Low data rate Mobile Earth Stations (MES)
except aeronautical mobile satellite earth stations,
operating in the 11/12/14 GHz frequency bands;
Harmonised Standard for access to radio spectrum

Reference
REN/SES-00428

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Foreword

This draft Harmonised European Standard (EN) has been produced by ETSI Technical Committee Satellite Earth Stations and Systems (SES), and is now submitted for the combined Public Enquiry and Vote phase of the ETSI Standardisation Request deliverable Approval Procedure (SRdAP).

The present document has been prepared under the Commission's standardisation request C(2015) 5376 final [i.3] to provide one voluntary means of conforming to the essential requirements of Directive 2014/53/EU on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC [i.6].

Once the present document is cited in the Official Journal of the European Union under that Directive, compliance with the normative clauses of the present document given in table A.1 confers, within the limits of the scope of the present document, a presumption of conformity with the corresponding essential requirements of that Directive and associated EFTA regulations.

Proposed national transposition dates	
Date of latest announcement of this EN (doa):	3 months after ETSI publication
Date of latest publication of new National Standard or endorsement of this EN (dop/e):	6 months after doa
Date of withdrawal of any conflicting National Standard (dow):	18 months after doa

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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Introduction

The present document is developed by ETSI and is designed to fit in a modular structure to cover all radio equipment under the RE Directive [i.6].

Figure 1: Void

The present document is based on ETSI TBR 027 [i.2].

The present document may not cover those cases where a potential source of interference which is producing individually repeated transient phenomena or a continuous phenomenon interference is present, e.g. a radar or broadcast site in the near vicinity. In such a case it is necessary to use special protection applied to either the source of interference, or the interfered part or both.

The present document does not contain any requirement, recommendation or information about the installation of the MES.

The determination of the parameters of the user earth stations using a given geostationary satellite for the protection of the spectrum allocated to that satellite, is considered to be under the responsibility of the satellite operator or the satellite network operators.

The present document is therefore intended to cover the provisions of Directive 2014/53/EU [i.6] (RE Directive) article 3.2 which states that "….*radio equipment shall be so constructed that it both effectively uses and supports the efficient use of radio spectrum in order to avoid harmful interference*".

As a consequence, the present document includes both transmitting and receiving parameters aiming to maximize the efficient use of radio spectrum.

1 Scope

The present document specifies technical characteristics and methods of measurements for Mobile Earth Stations (MES), except aeronautical mobile earth stations, which have the following characteristics:

- The MES are operating in one or more frequency ranges, or parts thereof, of the Fixed Satellite Service (FSS):
  - 10,70 GHz to 11,70 GHz (space to Earth);
  - 12,50 GHz to 12,75 GHz (space to Earth);
  - 14,00 GHz to 14,25 GHz (Earth to space).

The MES is designed to transmit to the Satellite in the 14,00 GHz to 14,25 GHz band and receive from the satellite in the bands 10,70 GHz to 11,70 GHz and 12,50 GHz to 12,75 GHz. The MES does not cause harmful interference to primary services in the band (including the Fixed Satellite Service (FSS)) and does not claim protection from harmful interference from those services.

- The MES may be either:
  - a Land Mobile Earth Station (LMES); and/or
  - a Maritime Mobile Earth Station (MMES) not providing those distress and safety functions required by the International Maritime Organization (IMO).
- These LMESs could be either vehicle mounted or portable equipment.
- These MMESs are installable equipment on ships.
- The MES could consist of a number of modules including a keyboard interface to the user.
- The MES use linear polarization.
- The MES operate through a geostationary satellite at least 3° away from any other geostationary satellite operating in the same frequency band and covering the same area.
- The antenna of the MES may be omnidirectional or directional with a means of tracking the satellite.
- The MES are operating as part of a satellite network used for the distribution and/or exchange of information between users.
- The MES are controlled and monitored by a Network Control Facility (NCF). The NCF is outside the scope of the present document.

The present document applies to the MES with its ancillary equipment and its various terrestrial ports, and when operated within the boundary limits of the operational environmental profile defined for its intended use and when installed, following the indications contained in the user documentation.

The present document is intended to cover the provisions of Directive 2014/53/EU [i.6] (RE Directive) article 3.2, which states that "… *radio equipment shall be so constructed that it both effectively uses and supports the efficient use of radio spectrum in order to avoid harmful interference*".

NOTE: The relationship between the present document and essential requirements of article 3.2 of Directive 2014/53/EU [i.6] is given in Annex A.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] Void.

[2] CISPR 16-1-1:2019: "Specification for radio disturbance and immunity measuring apparatus and methods - Part 1-1: Radio disturbance and immunity measuring apparatus".

[3] CISPR 16-1-5:2014: "Specification for radio disturbance and immunity measuring apparatus and methods - Part 1-5: Radio disturbance and immunity measuring apparatus - Antenna calibration sites and reference test sites for 5 MHz to 18 GHz".

[4] Void.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] Void.

[i.2] ETSI TBR 027: "Satellite Earth Stations and Systems (SES); Low data rate Land Mobile satellite Earth Stations (LMES) operating in the 11/12/14 GHz frequency bands".

[i.3] Commission Implementing Decision C(2015) 5376 final of 4.8.2015 on a standardisation request to the European Committee for Electrotechnical Standardisation and to the European Telecommunications Standards Institute as regards radio equipment in support of Directive 2014/53/EU of the European Parliament and of the Council.

[i.4] ETSI EG 203 336 (V1.2.1) (2020-05): "Guide for the selection of technical parameters for the production of Harmonised Standards covering article 3.1(b) and article 3.2 of Directive 2014/53/EU".

[i.5] ETSI TR 100 028-1 (V1.4.1): "Electromagnetic compatibility and Radio spectrum Matters (ERM); Uncertainties in the measurement of mobile radio equipment characteristics; Part 1".

[i.6] Directive 2014/53/EU of the European Parliament and of the Council of 16 April 2014 on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC (RE Directive).

[i.7] ERC Recommendation 74-01 (2021): "Unwanted emissions in the spurious domain".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in the Directive 2014/53/EU [i.6] and the following apply:

carrier-off state: state in which the MES is not transmitting a carrier

carrier-on state: state in which the MES is transmitting a carrier

Control Channel (CC): channel or channels by which MES receive control information from the NCF of their network

environmental profile: range of environmental conditions under which equipment shall operate

Externally Mounted Equipment (EME): modules of the Installable Equipment (IE) which are intended to be mounted externally to the vehicle as defined for the intended use of the MES

Installable Equipment (IE): equipment intended to be fitted to a vehicle

NOTE: An IE may consist of one or several interconnected modules.

Internally Mounted Equipment (IME): modules of the IE which are not provided with the EUT as EME are defined as IME

matched impedance: typically a dummy load inserted in the open port of the equipment to avoid open interface which may distort the measurements

NOTE: Its value is related to the intended use of the MES.

nominal antenna diameter: diameter of the antenna defined for the intended use of the MES

NOTE 1: An antenna with a nominal antenna diameter D has the same technical characteristics as a circular aperture antenna of diameter D. For non circular aperture antenna, the nominal antenna diameter is defined for the intended use of the MES.

NOTE 2: Network operators might request antennas of a certain diameter. Then an antenna that is compliant with the requirement for nominal antenna diameter equal to the requested antenna diameter can be used.

nominated bandwidth: bandwidth of the MES radio frequency transmission is nominated by its intended use

NOTE: The nominated bandwidth is wide enough to encompass all spectral elements of the transmission which have a level greater than the specified unwanted emissions limits. The nominated bandwidth is wide enough to take account of the transmit carrier frequency stability. The nominated bandwidth is within the transmit frequency band within which the MES operates.

Portable Equipment (PE): self-contained, free standing and portable

NOTE: Normally, a PE would consist of a single module, but may consist of several interconnected modules.

unwanted emissions: emissions including spurious and out of band emissions, falling inside and outside the nominated bandwidth depending on the intended use

wanted signal occupied Bandwidth (BW): occupied bandwidth of the wanted signal and it refers to the width of the signal spectrum 10 dB below the maximum in-band power density

3.2 Symbols

For the purposes of the present document, the following symbols apply:

dBsd	ratio expressed in decibels relative to the spectral density
dBc	ratio expressed in decibels relative to the EIRP of the unmodulated carrier
dBi	ratio of an antenna gain to the gain of an isotropic antenna, expressed in decibels
dBW	ratio of a power to 1 watt, expressed in decibels

dBpW ratio of a power to 1 picowatt, expressed in decibels

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

BW Wanted signal occupied Bandwidth
CC Control Channel
CISPR International Special Committee on Radio Interference
CMF Control and Monitoring Functions
EFTA European Free Trade Association
EIRP Equivalent Isotropically Radiated Power
EME Externally Mounted Equipment
EUT Equipment Under Test
FSS Fixed Satellite Service
IE Installable Equipment
IME Internally Mounted Equipment
IMO International Maritime Organization
LMES Land Mobile Earth Station
LNB Low Noise Block
MES Mobile Earth Station
MMES Maritime Mobile Earth Station
NCF Network Control Facility
OATS Open Area Test Site
PE Portable Equipment
RE Radio Equipment
RED Radio Equipment Directive
RF Radio Frequency
STE Special Test Equipment
TBR Technical Basis for Regulation

4 Technical requirement specifications

4.1 Environmental profile

The technical requirements of the present document apply under the environmental profile for operation of the equipment, which shall be in accordance with its intended use. The equipment shall comply with all the technical requirements of the present document at all times when operating within the boundary limits of the operational environmental profile defined by its intended use.

The environmental profile for operation of the equipment shall include the ranges of humidity, temperature and supply voltage.

4.2 Conformance requirements

4.2.1 Unwanted emissions outside the band 14,0 GHz to 14,25 GHz

4.2.1.1 Justification

Protection of terrestrial and satellite services from emissions caused by MES outside the band 14,00 GHz to 14,25 GHz, to meet the essential requirements as referred to in the scope of the present document.

4.2.1.2 Specification

The unwanted emissions in the measurement bandwidth and in all directions from the MES outside the band 14,00 GHz to 14,25 GHz, within which the MES is designed to operate, shall be below the following limits:

1) The unwanted emissions over the frequency range 30 MHz to 1 000 MHz shall not exceed the limits in table 1a for LMESs and table 1b for MMESs.

Table 1a: Limits of LMESs unwanted emissions up to 1 000 MHz at a measuring distance of 10 m in a 120 kHz bandwidth

Frequency (MHz)	Quasi-peak limits (dB(µV/m))
30 to 230	30
230 to 1 000	37

Table 1b: Limits of MMESs unwanted emissions up to 1 000 MHz at a measuring distance of 10 m in a 120 kHz bandwidth

Frequency (MHz)	Quasi-peak limits (dB(µV/m))
30 to 156	30
156 to 165	14 (see note)
165 to 230	30
230 to 1 000	37
NOTE: In the band 156 MHz to 165 MHz, a 9 kHz bandwidth apply.	

NOTE: The above limits are consistent with ERC Recommendation 74-01 [i.7].

The lower limit shall apply at the transition frequency.

2) The unwanted emissions Equivalent Isotropically Radiated Power (EIRP) above 1 000 MHz, in the measurement bandwidth and in all directions shall not exceed the limits given in table 2.

Table 2: Limits of unwanted emissions above 1 000 MHz and outside the band 14,00 GHz to 14,25 GHz

Frequency range	Carrier-on		Carrier-off	
(MHz)	EIRP limit (dBpW)	Measurement bandwidth (kHz)	EIRP limit (dBpW)	Measurement bandwidth (kHz)
1 000 to 1 525	49	100	48	100
1 525 to 1 559	49	100	17	3
1 559 to 3 400	49	100	48	100
3 400 to 10 700	55	100	48	100
10 700 to 21 200	61	100	54	100
21 200 to 40 000	67	100	60	100

The lower limits shall apply at the transition frequencies.

4.2.1.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.1.

4.2.2 Unwanted emissions within the band 14,00 GHz to 14,25 GHz

4.2.2.1 Justification

Protection of the primary services operating in the 14,00 GHz to 14,25 GHz frequency band, to meet the essential requirements as referred to in the scope of the present document.

4.2.2.2 Specification

Specification 1: Carrier-on state
The EIRP spectral density of the unwanted emissions shall not exceed 4 - 10 log N dBW/100 kHz in the 14,00 GHz to 14,25 GHz band and outside the nominated bandwidth.

N is the maximum number of MES which are expected to transmit simultaneously on the same frequency for the intended use of the MES. This number shall not be exceeded for more than 0,01 % of the time. The value of N and the operational conditions of the system shall be defined for the intended use of the MES.

Specification 2: Carrier-off state
The EIRP spectral density of any emission in the 14,00 GHz to 14,25 GHz band shall not exceed -21 dBW/100 kHz.

4.2.2.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.2.

4.2.3 Off-axis EIRP emissions density in the nominated bandwidth

4.2.3.1 Justification

Protection of other satellite systems which use the same frequency band, to meet the essential requirements as referred to in the scope of the present document.

4.2.3.2 Specification

For directional antennas the maximum EIRP in any 40 kHz band from any MES in any direction ϕ degrees from the antenna main beam axis shall not exceed the following limits within 3° of the geostationary orbit:

$33 - 25 \log (\phi + \delta\phi) - 10 \log (K)$ dBW/40 kHz where $2{,}5° \leq \phi + \delta\phi \leq 7{,}0°$;

$+12 - 10 \log (K)$ dBW/40 kHz where $7{,}0° < \phi + \delta\phi \leq 9{,}2°$;

$36 - 25 \log (\phi + \delta\phi) - 10 \log (K)$ dBW/40 kHz where $9{,}2° < \phi + \delta\phi \leq 48°$;

$-6 - 10 \log (K)$ dBW/40 kHz where $48° < \phi + \delta\phi \leq 180°$;

where ϕ is the angle, in degrees, between the main beam axis and the direction considered. The value of $\delta\phi$ is equal to either:

a) the rms antenna tracking accuracy; or

b) twice the static rms antenna pointing accuracy;

whichever is the larger.

The rms antenna tracking accuracy or the antenna pointing accuracy shall be defined for the intended use of the MES and provided with the EUT documentation.

K is the power density ratio between the fully loaded system (maximum number of simultaneous carriers transmitting in the defined intended use where the MES will operate and provided with the EUT documentation) and a single MES measured in a 40 kHz bandwidth

The value of K, all technical characteristics and the operational conditions defined for its intended use shall be entered in the test report.

These limits apply over the range of latitudes and for the geostationary orbital arc as defined for the intended use of the MES.

For non-directional antennas, the maximum EIRP per 40 kHz in any direction shall not exceed:

-6 - 10 log (K) dBW/40 kHz;

where K is as defined above. Because of this requirement for non directional antennas, the below requirements in clause 4 do not apply to them.

The Off-axis EIRP emissions shall not exceed the levels derived from this specification for 95 % of the peaks.

4.2.3.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.3, to meet the essential requirements as referred to in the scope of the present document.

4.2.4 Control and Monitoring Functions (CMF)

4.2.4.0 General

The following minimum set of CMF shall be implemented in MES in order to minimize the probability that they may originate unwanted transmissions that may give rise to harmful interference to other systems.

Under any fault condition when the MES transmissions are being suppressed the limits for carrier-off state shall apply.

4.2.4.1 Processor monitoring

4.2.4.1.1 Justification

To ensure that the MES can suppress transmissions in the event of a processor sub-system failure. to meet the essential requirements as referred to in the scope of the present document.

4.2.4.1.2 Specification

The MES shall incorporate a processor monitoring function for each of its processors involved in the manipulation of traffic and in control and monitoring functions.

The processor monitoring function shall detect any failure of the processor hardware and software.

No later than one second after any fault condition occurs, the MES shall enter the carrier-off state until the processor monitoring function has determined that all fault conditions have been cleared.

4.2.4.1.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.4.2.

4.2.4.2 Transmit subsystem monitoring

4.2.4.2.1 Justification

To ensure the correct operation of the transmit frequency generation sub-system, and to inhibit transmissions should the sub-system fail, to meet the essential requirements as referred to in the scope of the present document.

4.2.4.2.2 Specification

The MES shall monitor the operation of its transmit frequency generation sub-system.

No later than 5 seconds after any fault condition of the transmit frequency generation sub-system occurs, the MES shall enter the carrier-off state until the transmit sub-system monitoring function has determined that all fault conditions have been cleared.

4.2.4.2.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.4.3.

4.2.4.3 Power-on/Reset

4.2.4.3.1 Justification

To demonstrate that the MES achieves a controlled non-transmitting state following the powering of the unit or the occurrence of a reset made by a local operator when this function is implemented, to meet the essential requirements as referred to in the scope of the present document.

4.2.4.3.2 Specification

During and following "power on" or a manual reset when this function is implemented, the MES shall remain in the carrier-off state.

4.2.4.3.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.4.4.

4.2.4.4 Control Channel (CC) reception

4.2.4.4.1 Justification

To ensure that the MES cannot transmit unless it correctly receives the CC messages from the NCF, to meet the essential requirements as referred to in the scope of the present document.

4.2.4.4.2 Specification

a) Without correct reception of the CC messages from the NCF, the MES shall remain in the carrier-off state.

b) The MES shall enter the carrier-off state immediately after a period not exceeding 30 seconds without correct reception of the CC messages from the NCF.

4.2.4.4.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.4.5.

4.2.4.5 Network control commands

4.2.4.5.1 Justification

These requirements ensure that the MES is capable of:

a) retaining a unique identification in the network and transmitting it upon reception of an appropriate request;

b) receiving commands from the NCF through its CC and executing those commands.

and allow to meet the essential requirements as referred to in the scope of the present document.

4.2.4.5.2 Specification

The MES shall hold, in non-volatile memory, its unique identification code in the network.

The MES shall be capable of receiving through its CC dedicated commands (addressed to the MES) from the NCF, and which contain:

- transmission enable commands;

- transmission disable commands;
- identification request.

Once a transmission enable command is received the MES is authorized to transmit.

After power-on or reset the MES shall remain in the carrier-off state until it receives a transmission enable command. For systems where no transmission enable command is expected after power-on or reset the MES may only transmit initial bursts (see clause 4.2.4.6).

Once a transmission disable command is received, within 1 second the MES shall enter and shall remain in the carrier-off state until the transmission disable command is superseded by a subsequent transmission enable command.

The MES shall be capable of transmitting its identification code upon reception of an identification request.

4.2.4.5.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.4.6.

4.2.4.6 Initial burst transmission

4.2.4.6.1 Justification

To limit disturbance to other services.

4.2.4.6.2 Specification

For systems where no transmission enable command is expected after power-on or reset the MES may transmit initial bursts:

a) the transmission of the initial burst shall not exceed 1 % of the time;

b) each burst shall not last more than one second.

4.2.4.6.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.4.7.

4.2.5 Receive antenna off-axis gain pattern

4.2.5.1 Justification

To protect the wanted signals from interference from terrestrial services and from other satellite services. The receive antenna shall be directional, to meet the essential requirements as referred to in the scope of the present document.

4.2.5.2 Specification

The maximum antenna gain of each of the co-polarized components in any direction φ degrees from the antenna main beam axis shall not exceed the following limits:

$G = 32 - 25 \log \varphi$ dBi for $\varphi_{min} \leq \varphi < 48°$

$G = -10$ dBi for $48° \leq \varphi \leq 85°$

$G = 0$ dBi for $85° \leq \varphi \leq 180°$

where:

$\varphi_{min} = 1°$ or $100\ \lambda/D$ degrees, whichever is the greater, for $D/\lambda \geq 50$.

$\varphi_{min} = 2°$ or $114\ (D/\lambda)^{-1,09}$ degrees, whichever is the greater, for $D/\lambda < 50$.

D is the nominal diameter of the antenna.

In addition the maximum antenna gain of each of the cross-polarized components in any direction ϕ degrees from the antenna main beam axis shall not exceed the following limits:

$G_x(\varphi) = 23 - 20 \log \varphi$ dBi for $\varphi_r \leq \varphi \leq 7°$

where φ_r is equal to 1° or 100 λ/D, whichever is greater

4.2.5.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.5.

4.2.6 Blocking performance

4.2.6.1 Justification

To prevent high power signals outside the receive frequency band from blocking the reception of signals inside the receive frequency band, and to meet the essential requirements as referred to in the scope of the present document.

4.2.6.2 Specification

Receiver blocking is characterized here through gain compression for a signal inside the receive frequency band that is caused by another signal outside the receive frequency band at high power. The level of the other signal is compared to the level of a signal inside the receive frequency band that would cause the same gain compression.

Receiver blocking rejection at a particular frequency is defined as the level of a second signal at this frequency that causes a certain gain compression to a first signal inside the receive frequency band, minus the level of a second signal at a frequency inside the receive frequency band that causes the same gain compression.

The first signal shall be the wanted modulated signal as defined for the intended use, tuned at the centre frequency of the receive frequency band. The second signal shall be a CW signal with a level of 30 dB above the reference sensitivity level and shall not cause a gain compression for the first signal higher than 1 dB.

The rejection shall comply with table 3.

Table 3: Receiver blocking rejection

Frequency	Minimum rejection
below 9 GHz	20 dB
9 to 10 GHz	10 dB
14 to 16 GHz	10 dB
above 16 GHz	20 dB

4.2.6.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.6.

4.2.7 Adjacent Signal Selectivity

4.2.7.1 Justification

To enable reception of a wanted signal in presence of other signals on adjacent frequencies which are transmitted with high EIRP density from the same satellite orbital position and to meet the essential requirements as referred to in the scope of the present document.

NOTE: The power level of signals transmitted from the same orbital position are under control of the satellite operator. Signals transmitted from an adjacent orbital position that is not near-by are suppressed by the antenna gain pattern.

4.2.7.2 Specification

Adjacent Signal Selectivity is a measure of a receiver's ability to receive a signal at its assigned channel frequency in the presence of an adjacent signal at a given frequency offset from the centre frequency of the assigned channel. The adjacent signal shall occupy the same bandwidth as the wanted signal as per the modulation defined for the intended use. Frequency offset and relative power level of the adjacent signal compared to the wanted signal shall take the value given in table 4. BW is the wanted signal occupied bandwidth.

Table 4: Adjacent signal frequency and Power level

Signal	Centre frequency offset from wanted signal	Power level relative to wanted signal
Adjacent Carrier	BW	7 dBsd

The decrease in the required signal to noise ratio in the presence of an adjacent signal shall be no more than 0,5 dB.

4.2.7.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.7.

5 Testing for compliance with technical requirements

5.1 Environmental conditions for testing

Tests defined in the present document shall be carried out at representative points within the boundary limits of the operational environmental profile defined by its intended use, which, as a minimum, shall be that specified in the test conditions contained in the present document.

Where technical performance varies subject to environmental conditions, tests shall be carried out under a sufficient variety of environmental conditions as specified in the present document to give confidence of compliance for the affected technical requirements.

5.2 Essential radio test suites

The essential radio test suites for a MES are given in clause 6.

6 Test methods

6.0 General

To enable the performance tests to be carried out the use of Special Test Equipment (STE), made available with the equipment under test, may be necessary. Since this test equipment will be specific for the particular system, it is not possible to provide detailed specifications in the present document. However, the following baseline is provided:

- if the MES requires to receive a modulated carrier from the satellite in order to transmit, then special test arrangements are required to simulate the satellite signal, thus enabling the MES to transmit allowing measurement of transmission parameters;
- any characteristic of these special test arrangements which may have direct or indirect effects on the parameters to be measured shall be clearly defined for the intended use of the MES.

All tests with carrier-on shall be undertaken with the transmitter operating at full power and with the maximum transmit burst rate where applicable.

All technical characteristics and operational conditions, as defined for the intended use of the MES, shall be entered in the test report.

6.1 Unwanted emissions outside the band 14,00 GHz to 14,25 GHz

6.1.1 General

For purpose of the test, the Equipment Under Test (EUT) comprises:

a) for IE:

- the EME;
- the IME;
- interconnecting cables between IME and EME units as supplied by the manufacturer;
- the necessary power supply cables and any other cable ensuring a proper functioning of the terminal.

b) for PE:

- for a single module PE, the module itself with any deployable parts in their normal operating configuration;
- for a multiple module PE, all such modules with all necessary interconnecting cables of lengths as normally provided with the EUT; again any deployable parts should be in their normal operating configuration.

For measurements up to 1 000 MHz the distance between the EUT and the measuring antenna shall be 10 m.

6.1.2 Test site

This test shall be performed using the test method and test arrangement specified in the following clauses.

The tests can be carried out in an Open Area Test Site (OATS) environment or in an anechoic chamber or in a semi-anechoic chamber as they lead to comparable test results for establishing compliance. Ambient noise levels shall be at least 6 dB below the applicable unwanted emissions limit. The preferred test site is the semi-anechoic chamber.

An open area test site shall be flat, free of overhead wires and nearby reflecting structures, permitting aerial placement at the specified measuring distance and provide separation between aerial, test unit and reflecting structures, according to the specification of CISPR 16-1-5 [3].

For both the open area test site and the semi-anechoic chamber a metal ground plane shall be inserted on the natural ground plane and it shall extend at least 1 m beyond the perimeter of the EUT at one end and at least 1 m beyond the measurement antenna at the other end.

6.1.3 Test method

6.1.3.0 General

For IE, the EUT shall be installed with a separation of about 0,5 m between the IME and the EME, the maximum length connection cable specified for its intended use shall be installed. The height of the cable shall be between 0,5 m and 1 m. The cable shall be maintained in that position by non-metallic means. The EME shall be set, in its normal operating configuration on a non-metallic table at a height between 0,5 m and 1 m. The IME shall be set on a non-metallic table at a height of 0,8 m for tests up to 1 000 MHz and between 0,5 m and 1 m for tests above 1 000 MHz. Any associated equipment, e.g. portable computer or data terminal if required for operation of the MES, shall be placed next to, and at the same height as, the IME.

For PE, the equipment shall be arranged in its normal operating configuration as defined for the intended use of the MES on a non-metallic table at a height between 0,5 m and 1 m.

The EUT shall be terminated with matched impedances (e.g. dummy load) at the terrestrial ports as defined for the intended use of the MES, provided in the EUT documentation and when there is no associated equipment connected to such ports.

For frequencies up to 80 MHz, the measuring antenna shall be a balanced dipole which has a length equal to the 80 MHz resonant length and shall be matched to the feeder by a suitable transforming device. Measurements with broad band antennas is also possible provided that the test site has been calibrated according to the requirements of CISPR 16-1-1 [2].

For frequencies between 80 MHz and 1 000 MHz, the measuring antenna shall be a balanced dipole which shall be resonant in length. Measurements with broad band antennas is also possible provided that the test site has been calibrated according to the requirements of CISPR 16-1-1 [2].

For frequencies above 1 000 MHz, the antenna shall be a horn radiator of known gain/frequency characteristics. When used for reception, the antenna and any associated amplification system shall have an amplitude/frequency response within ±2 dB of the combined calibration curves across the measurement frequency range considered for the antenna. The antenna is mounted on a support capable of allowing the antenna to be used in either horizontal or vertical polarization and at the specified height.

For tests up to 1 000 MHz, the receive test equipment shall be a measuring receiver.

For tests above 1 000 MHz, the receive test equipment shall be a spectrum analyser.

6.1.3.1 Receive test equipment

6.1.3.1.1 Measuring receiver for measurements up to 1 000 MHz

Measuring receivers shall conform to the following characteristics:

- the response to a constant amplitude sine wave signal shall remain within ±1 dB across the frequency range of interest;
- quasi-peak detection shall be used in a -6 dB bandwidth of 120 kHz;
- the receiver shall be operated at more than 1 dB below the compression point during tests/measurements.

6.1.3.1.2 Spectrum analyser for measurements above 1 000 MHz

The spectrum analyser resolution bandwidth shall be set to the specified measuring bandwidth or as close as possible. If the resolution bandwidth is different from the specified measuring bandwidth, linear bandwidth correction shall be performed for noise-like wideband signals. The measuring system shall be capable of detecting signals at least 6 dB below the applicable unwanted emissions limit.

6.1.4 Procedure

6.1.4.1 Test arrangements

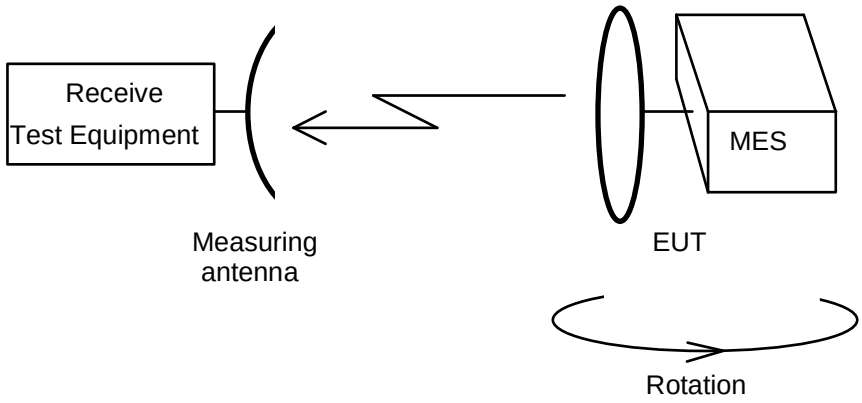


Figure 2: Test arrangement - unwanted emissions measurement, first axis

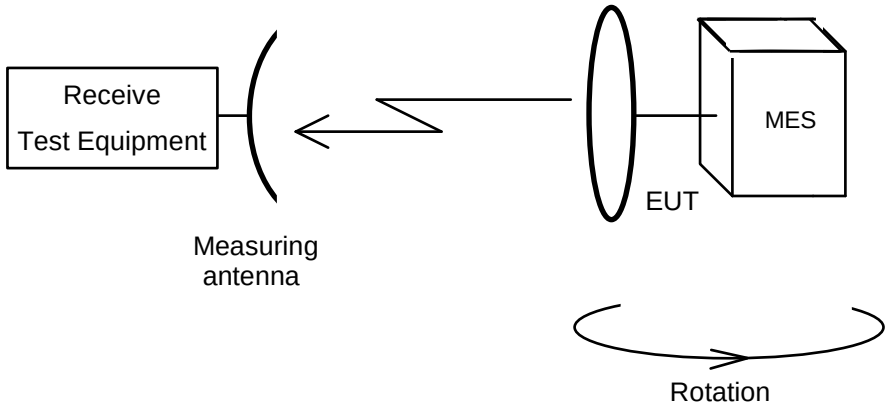


Figure 3: Test arrangement - unwanted emissions measurement, second axis

The EUT shall be installed at a height between 0,5 m to 1 m.

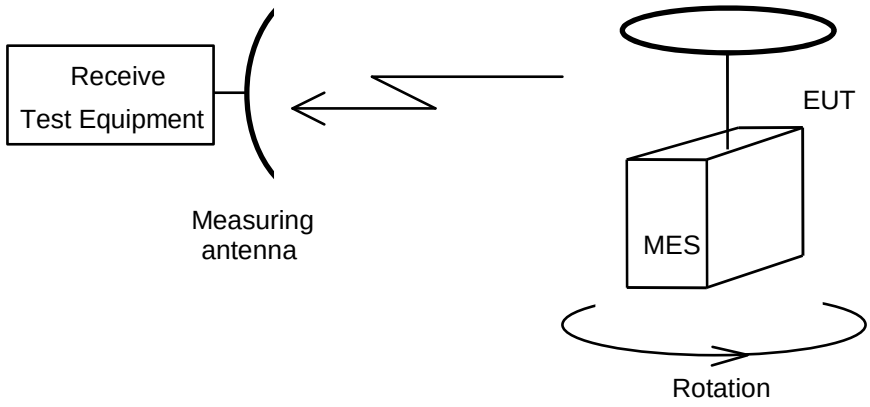


Figure 4: Test arrangement - unwanted emissions measurement, third axis

6.1.4.2 Up to 1 000 MHz

a) The test arrangement shall be as shown in figure 2 with the measuring receiver installed. EUTs with adjustable antennas shall have the antenna boresight axis in the plane of rotation. The measuring antenna boresight axis shall coincide with the plane of rotation of the boresight of the EUT.

b) The EUT shall be in the carrier-on state with the carrier at the lowest possible centre frequency.

c) The EUT shall be rotated through 360° whilst unwanted emissions are measured in frequency and amplitude, over the frequency range 30 MHz to 1 000 MHz. The frequency and amplitude of each signal shall be noted.

d) The measurements shall be repeated with the measuring antenna in the opposite polarization and the signal levels similarly noted.

e) The tests in c) and d) above shall be repeated with the EUT carrier at the highest possible centre frequency.

f) The tests in c) and d) above shall be repeated with the carrier-off.

g) The tests in b) to f) above shall be repeated with the EUT turned so that its axis of rotation is orthogonal to that of the first case, as shown in figure 3. The EUT antenna boresight axis shall remain in the plane of rotation.

h) The tests in b) to f) above shall be repeated with the EUT turned so that its axis of rotation is mutually orthogonal to those of the first two cases, as shown in figure 4. The EUT antenna boresight axis shall be perpendicular to the plane of rotation.

6.1.4.3 Above 1 000 MHz

6.1.4.3.0 General

For MES equipment for which measurements at the antenna flange are not possible or not foreseen by the intended use, the EUT shall be tested with the antenna. The test shall be performed in two stages:

- procedure a): identification of the significant frequencies of the radiated unwanted emissions;
- procedure b): measurement of radiated power levels of identified unwanted emissions.

For MES equipment for which measurements at the antenna flange are possible and foreseen by its intended use, the EUT shall be tested without the antenna. The test shall be performed in three stages:

- procedure a): identification of the significant frequencies of the radiated unwanted emissions;
- procedure b): measurement of radiated power levels of identified unwanted emissions;
- procedure c): measurement of conducted unwanted emissions.

6.1.4.3.1 Identification of the significant frequencies of the radiated unwanted emissions

a) The test arrangement shall be as shown in figure 2 with the spectrum analyser installed. EUTs with adjustable antennas shall have the antenna boresight axis in the plane of rotation. The measuring antenna boresight axis shall coincide with the plane of rotation of the boresight of the EUT.

b) The EUT shall be in the carrier-on state with the carrier at the lowest possible centre frequency.

c) The EUT shall be rotated through 360° whilst unwanted emissions are measured in frequency and amplitude, over the frequency range 1 000 MHz to 40 GHz. The frequency and amplitude of each signal shall be noted.

d) The measurements shall be repeated with the measuring antenna in the opposite polarization and the signal levels similarly noted.

e) The tests in c) and d) above shall be repeated with the EUT carrier at the highest possible centre frequency.

f) The tests in c) and d) above shall be repeated with the carrier-off.

g) The tests in b) to f) above shall be repeated with the EUT turned so that its axis of rotation is orthogonal to that of the first case, as shown in figure 3. The EUT antenna boresight axis shall remain in the plane of rotation.

h) The tests in b) to f) above shall be repeated with the EUT turned so that its axis of rotation is mutually orthogonal to those of the first two cases, as shown in figure 4. The EUT antenna boresight axis shall be perpendicular to the plane of rotation.

6.1.4.3.2 Measurement of radiated power levels of identified spurious radiation

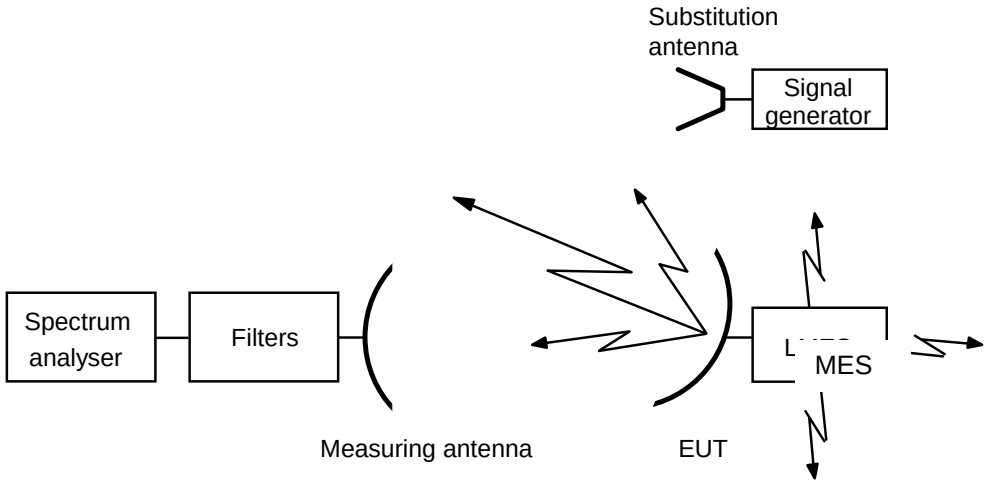


Figure 5: Test arrangement - unwanted emissions measurement for an EUT with antenna

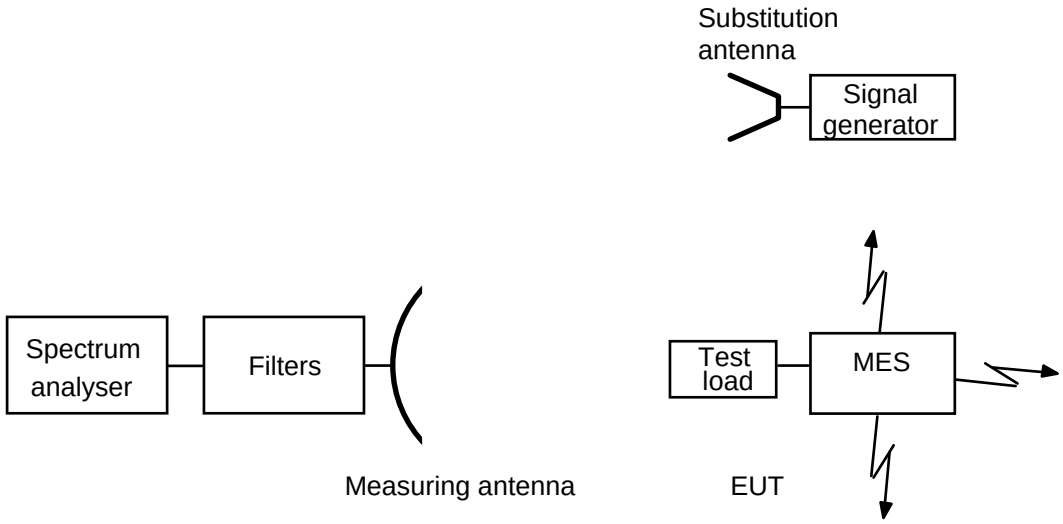


Figure 6: Test arrangement - unwanted emissions measurement for an EUT without antenna

a) The test arrangement shall be as shown in figures 5 or 6.

b) The EUT shall be in the carrier-on state with the modulated carrier centre frequency as appropriate to the previously identified unwanted emission.

c) The measuring antenna shall be adjusted in height and the EUT rotated for a maximum response on the associated spectrum analyser at each unwanted emission previously identified, this response level shall be noted.

d) For the test arrangement in figure 2 the investigation shall be repeated with the measuring antenna in the opposite polarization and the response level similarly noted.

e) The EUT shall be replaced by the substitution antenna to which is connected a signal generator. The main beam axes of the measuring and substitution antennas shall be aligned.

f) The substitution and measuring antennas shall be aligned in the polarization which produced the larger response between the EUT and the test antenna, in steps c) and d).

g) The output of the generator shall be adjusted so that the received level is identical to that of the previously noted largest spurious radiation.

h) The output level of the signal generator shall be noted. The EIRP of the unwanted emission is the sum, in dB, of the signal generator output plus the substitution antenna isotropic gain minus the interconnection cable loss.

6.1.4.3.3 Measurement of conducted unwanted emissions at the antenna flange

6.1.4.3.3.1 Test site

There are no requirements for the test site to be used for this test.

6.1.4.3.3.2 Procedure

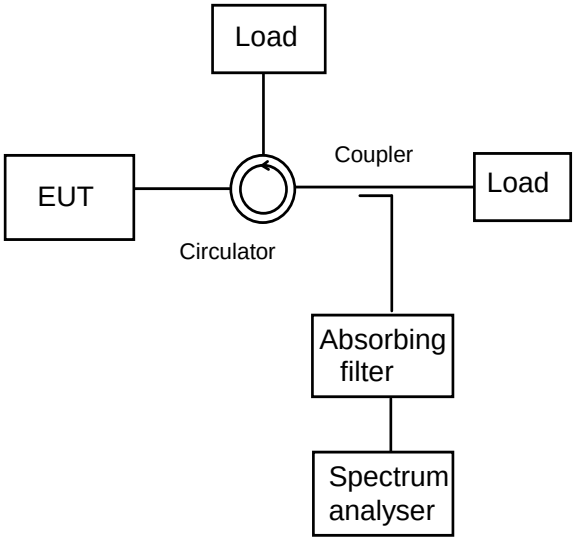


Figure 7: Test arrangement - conducted unwanted emissions

a) The test arrangement shall be as shown in figure 7. The absorbing filter shall be tuned to the transmit carrier frequency, in order to protect the spectrum analyser while ensuring the necessary measurement accuracy, particularly close to the carrier.

b) The frequency range 1 000 MHz to 40 GHz shall be investigated for unwanted emissions whilst in the carrier-on state with the carrier at maximum power and normally modulated. The power density of the unwanted emissions identified shall be measured.

c) To obtain the maximum EIRP, the maximum antenna gain measured at the frequency of the identified unwanted emission shall be added to the measured power density and any correction or calibration factors summated with the result. If foreseen by the intended use, it shall be acceptable that the maximum antenna gain measured under clause 6.3.3.2 is used in place of the maximum antenna gain at the frequency of the identified unwanted emission.

d) The test shall be repeated in the carrier-off state.

6.2 Unwanted emissions within the band 14,00 GHz to 14,25 GHz

6.2.1 Test method

6.2.1.0 Introduction

For purpose of the test, the EUT comprises:

a) for IE:

- the EME;
- the IME;
- a connection cable between IME and EME unit;
- the necessary power supply cables and any other cable ensuring a proper functioning of the terminal;

b) for PE:

- for a single module PE, the module itself with any deployable parts in their normal operating configuration;
- for a multiple module PE, all such modules with all necessary interconnecting cables of lengths as normally provided with the equipment; again any deployable parts should be in their normal operating configuration.

6.2.1.1 General

For MES equipment for which measurements at the antenna flange are possible and foreseen by the intended use, the measurements shall be performed at the antenna flange.

For MES equipment for which measurements at the antenna flange are not possible or not foreseen for the intended use, the measurements shall be performed with a test antenna.

6.2.1.2 Method of measurement at the antenna flange

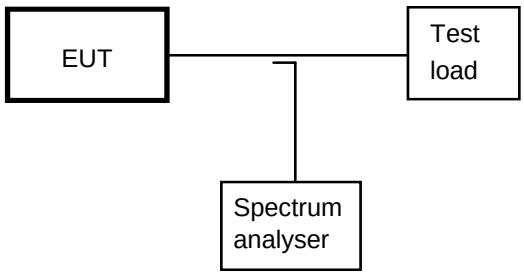


Figure 8: Test arrangement - on-axis unwanted emissions measurements at the antenna flange

a) The EUT shall be connected to a test load as shown in figure 8.

b) The frequency range 14,00 GHz to 14,25 GHz shall be investigated. For carrier-on state, the measurement shall not be performed within the nominated bandwidth. The measurement shall be performed whilst transmitting a modulated carrier at maximum power. The carrier frequency shall be as close as possible to the lower limit of the operating frequency band of the EUT as defined for the intended use of the MES.

c) The spectrum analyser resolution bandwidth shall be set to the appropriate measuring bandwidth or as close as possible. If the resolution bandwidth is different from the specified measuring bandwidth, linear bandwidth correction shall be performed for noise-like wideband spurious radiation.

d) To obtain the on-axis unwanted emission power level that would be transmitted, the antenna isotropic transmit gain shall be added to any figure obtained in the above measurement and any correction or calibration factor summated with the result.

e) The antenna gain shall be as measured in clause 6.3.3.2.

f) The tests in b) to e) shall be repeated in the carrier-off state.

g) The tests in b) to f) shall be repeated with a transmit frequency as close to the upper limit of the operating frequency band of the EUT as possible.

6.2.1.3 Method of measurement with a test antenna

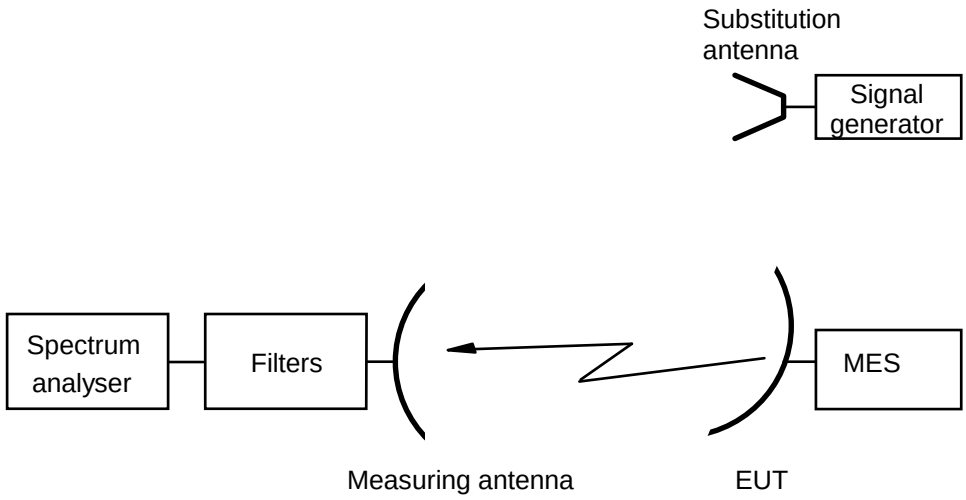


Figure 9: Test arrangement - on-axis unwanted emissions measurements with a measuring antenna

a) The test arrangement shall be as shown in figure 9.

b) The frequency range 14,00 GHz to 14,25 GHz shall be investigated. For carrier-on state, the measurement shall not be performed within the nominated bandwidth. The measurement shall be performed whilst transmitting a modulated carrier at maximum power. The carrier frequency shall be as close as possible to the lower limit of the operating frequency band of the EUT as foreseen by its intended use.

c) The spectrum analyser resolution bandwidth shall be set to the appropriate measuring bandwidth or as close as possible. If the resolution bandwidth is different from the specified measuring bandwidth, linear bandwidth correction shall be performed for noise-like wideband spurious radiation.

d) The EUT shall be installed such that the various units are placed in their normal operating positions relative to each other. Any interconnection cables shall be maintained by non-metallic means at a height between 0,5 m and 1,0 m.

e) The measuring antenna shall be adjusted in height and the EUT rotated for a maximum response on the associated spectrum analyser at each unwanted emission identified, this response level shall be noted.

f) The measuring antenna shall be adjusted in height and polarization and the EUT rotated for a maximum response on the associated spectrum analyser at each unwanted emission previously identified, this response level shall be noted.

g) The EUT shall be replaced by the substitution antenna to which is connected a signal generator. The main beam axes of the measuring and substitution antennas shall be aligned.

h) The substitution and measuring antennas shall be aligned in the polarization which produced the larger response between the EUT and the test antenna, in steps e) and f).

i) The output of the generator shall be adjusted so that the received level is identical to that of the previously noted largest unwanted emission.

j) The output level of the signal generator shall be noted. The EIRP of the on-axis unwanted emission is the sum, in dB, of the signal generator output plus the substitution antenna isotropic gain minus the interconnection cable loss.

k) The tests in d) to j) shall be repeated in the carrier-off state.

6.3 Off-axis EIRP emissions density in the nominated bandwidth

6.3.1 General

Conformance shall be determined from:

a) the measurement of the static rms antenna pointing accuracy;

b) the measurement of the off-axis EIRP.

For the measurement of the off-axis EIRP, the EUT shall be either with or without its antenna:

- For MES equipment for which measurements at the antenna flange are possible and foreseen by its intended use, the EUT shall be without the antenna. The test shall be performed in three stages:

  a) the measurement of the transmitter output power density (dBW/40 kHz);

  b) the measurement of the antenna transmit gain (dBi);

  c) the measurement of the antenna transmit radiation patterns (dBi).

- For MES equipment for which measurements at the antenna flange are not possible or not foreseen by its intended use, the EUT shall be fitted with the antenna. The test shall be performed in three stages:

  a) the measurement of the maximum EIRP density per 40 kHz ratio relative to the EIRP (dBc/40 kHz);

  b) the measurement of the maximum on-axis EIRP (dBW);

  c) the measurement of the antenna transmit radiation patterns (dBi).

6.3.2 Static rms antenna pointing accuracy

6.3.2.1 Method of measurement

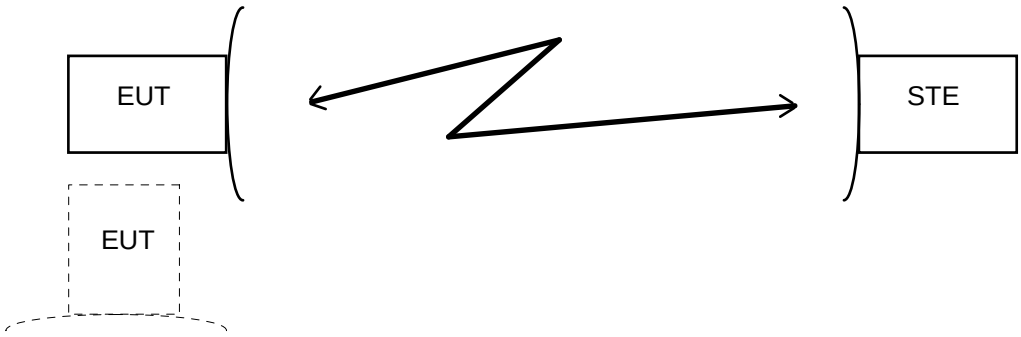


Figure 10: Test arrangement - static rms antenna pointing accuracy

a) The equipment shall be arranged as shown in figure 10 such that the two antennas are in the far field of each other with the EUT rotated away from the STE. The STE shall be switched-on and the EUT shall be switched-off.

b) The level of the signal radiated by the STE shall be adjusted so that the EUT receives a power density corresponding to the signal to noise ratio as defined for the intended use of the MES. This power density shall be representative of those expected for 95 % of the MES within the network and typically will be 2 dB above the edge of coverage figures for which the MES is designed to operate.

c) The EUT shall be switched-on and allowed to acquire the static antenna pointing position.

d) The static antenna pointing accuracy shall be measured and recorded. The method of measurement to be used shall be consistent for the intended use.

e) The static antenna pointing accuracy shall be monitored to ascertain if the pointing varies, if so the rms value of this pointing angle shall be taken.

f) The static antenna pointing accuracy shall be measured five times.

g) The EUT shall be switched-off whilst the EUT is rotated through at least 90° and for a minimum period of 1 s.

h) The tests in c) to f) shall be repeated.

i) The value of rms static antenna pointing accuracy shall be taken to be the largest value of the ten rms measurement results recorded.

6.3.3 Measurement of the off-axis EIRP without the antenna

6.3.3.1 Transmitter output power density

6.3.3.1.0 General

For purpose of the test, the EUT comprises all of the equipment noted up to the antenna flange:

a) for IE:

- the EME;
- the IME;
- a connection cable between IME and EME unit;
- the necessary power supply cables and any other cable ensuring a proper functioning of the terminal;

b) for PE:

- for a single module PE, the module itself with any deployable parts in their normal operating configuration;
- for a multiple module PE, all such modules with all necessary interconnecting cables of lengths as normally provided with the equipment; again any deployable parts should be in their normal operating configuration.

Where the EUT is so designed that it is not normally possible to make a direct connection to the feed at the antenna flange, or connecting point, the EUT should be equipped with a means of so doing specifically for the test MES.

6.3.3.1.1 Method of measurement

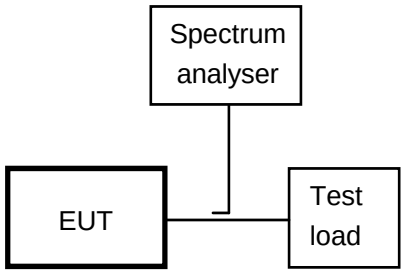


Figure 11: Test arrangement - transmit output power density measurement

a) The EUT shall be connected to a test load as shown in figure 11.

b) With the carrier being modulated by a pseudorandom bit sequence, the maximum power density supplied to the antenna flange shall be measured in dBW/40 kHz. The coupling factor of the test coupler at the test frequency and the attenuation of any necessary waveguide adapter shall be taken into account. The resolution bandwidth of the spectrum analyser shall be set as close as possible to the specified measuring bandwidth. If the resolution bandwidth is different from the specified bandwidth then linear bandwidth correction shall be performed.

6.3.3.2 Antenna transmit gain

6.3.3.2.1 General

For the purpose of the present document, the antenna transmit gain is defined as the ratio, expressed in decibels, of the power that would have to be supplied to the reference antenna, i.e. an isotropic radiator isolated in space, to the power supplied to the antenna being considered, so that they produce the same field strength at the same distance in the same direction. Unless otherwise specified the gain is for the direction of maximum radiation.

For the purpose of this test the EUT is defined as that part of the MES which comprises the antenna and its flange. The EUT includes an enclosure of equal weight/distribution to any electrical equipment normally housed within the antenna (see clause 6.3.3.1 regarding this requirement).

6.3.3.2.2 Test site

This test shall be performed using the method of measurement and test arrangement specified in the following clauses.

The tests can be carried out in an Open Area Test Site (OATS) environment or in a compact test range or in an anechoic chamber or in a semi-anechoic chamber as they lead to comparable test results for establishing compliance. Ambient noise levels shall be at least 6 dB below the applicable unwanted emissions limit. The preferred test site is the semi-anechoic chamber.

For both the open area test site and the semi-anechoic chamber a metal ground plane shall be inserted on the natural ground plane and it shall extend at least 1 m beyond the perimeter of the EUT at one end and at least 1 m beyond the measurement antenna at the other end.

6.3.3.2.3 Method of measurement

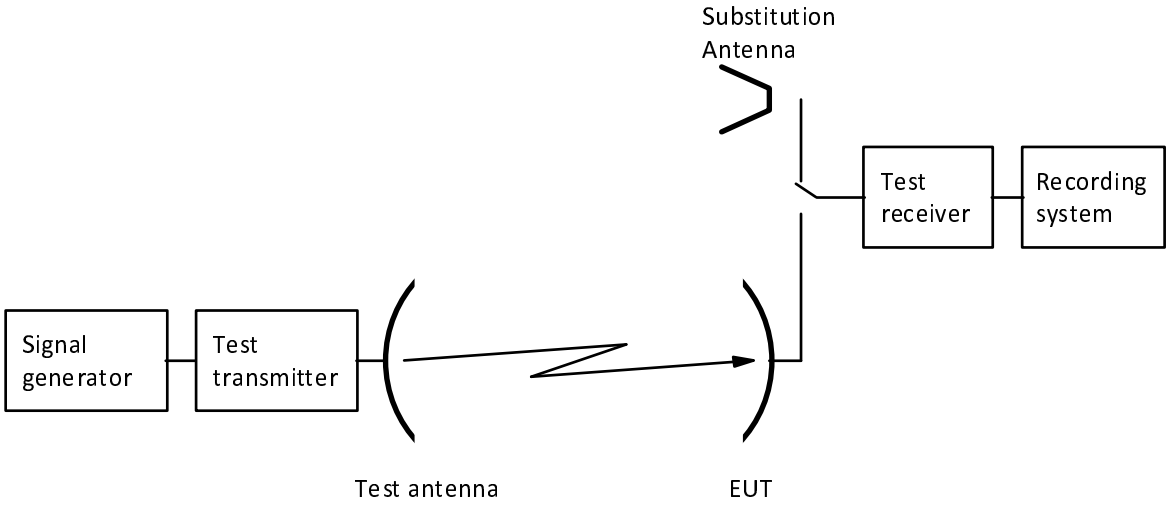


Figure 12: Test arrangement - antenna transmit gain measurement

a) The test arrangement shall be as shown in figure 12 with the EUT connected to the test receiver. A signal proportional to the angular position from the servo mechanism and the signal level from the test receiver shall be applied to the recording system.

b) The initial E-plane of the test signal radiated by the test transmitter through its antenna shall be horizontal. The EUT antenna main beam axis shall be aligned with the main beam axis of the test transmitter. The EUT antenna or its polarizer shall be rotated around its main beam axis such that its E-plane coincides with the E-plane of the test transmitter.

c) After any change of the E-plane of the test signal, the EUT antenna shall be rotated around its main beam axis such that its E-plane coincides with the E-plane of the test transmitter.

d) The frequency of the test signal shall be set 5 MHz above the bottom of the lowest frequency band as defined for the intended use of the MES.

e) The EUT shall be aligned to maximize the received signal and recording system adjusted to give the maximum reading on the chart.

f) The EUT shall be driven in azimuth in one direction through 10°.

g) The pattern measurement is then obtained by driving the EUT in azimuth back through boresight to 10° the other side with the plotter recording the results.

h) The EUT shall be replaced by the substitution antenna and the received signal level maximized.

i) This level shall be recorded on the recording system.

j) The substitution antenna shall be driven in azimuth as in f) and g).

k) The gain of the EUT shall be calculated from:

$$G_{EUT} = L_1 - L_2 + C$$

where:

- G_{EUT} is the gain of the EUT (dBi);
- L_1 is the level obtained with the EUT (dB);
- L_2 is the level obtained with the substitution antenna (dB);
- C is the calibrated gain of the substituted antenna at the test frequency (dBi).

l) The tests in e) to k) shall be repeated with the frequency changed to the middle of the lowest frequency band as defined for the intended use of the MES.

m) The tests in e) to k) shall be repeated with the frequency changed to 5 MHz below the top of the lowest frequency band as defined for the intended use of the MES.

n) The tests in d) to m) may be performed simultaneously.

o) The tests in c) to n) shall be repeated with the E-plane vertical.

p) The tests in c) to n) shall be repeated with the E-plane +45° to the horizontal plane.

q) The tests in c) to n) shall be repeated with the E-plane -45° to the horizontal plane.

r) The tests in b) to q) shall be repeated for all frequency bands as defined for the MES intended use.

6.3.3.3 Antenna transmit radiation patterns

6.3.3.3.1 General

For the purpose of the present document, the antenna transmit radiation patterns are diagrams relating field strength to direction relative to the pointing angle of the antenna at a constant large distance from the antenna.

For the purpose of this test the EUT is defined as that part of the MES which comprises the antenna and its flange. The antenna includes the parabolic reflector, feed, support struts and an enclosure of equal weight/distribution to any electrical equipment normally housed with the feed at the antenna focal point (see clause 6.3.3.1 regarding this requirement).

6.3.3.3.2 Test site

This test shall be performed using the following method of measurement and test arrangement as specified in the following clauses.

The tests can be carried out in an Open Area Test Site (OATS) environment or in a compact test range or in an anechoic chamber or in a semi-anechoic chamber as they can provide comparable results. Ambient noise levels shall be at least 6 dB below the applicable unwanted emissions limit. The preferred test site is a semi-anechoic chamber.

For both the open area test site and the semi-anechoic chamber a metal ground plane shall be inserted on the natural ground plane and it shall extend at least 1 m beyond the perimeter of the EUT at one end and at least 1 m beyond the measurement antenna at the other end.

6.3.3.3.3 Method of measurement

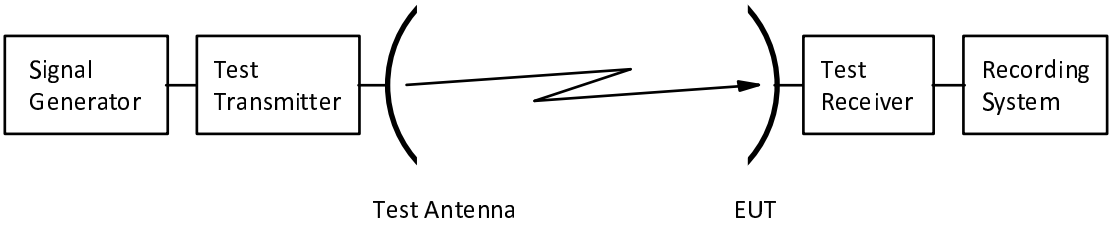


Figure 13: Test arrangement - antenna transmit radiation pattern measurement

a) The test arrangement shall be as shown in figure 13 with the EUT connected to the test receiver. A signal proportional to the angular position from the servo mechanism and the signal level from the test receiver shall be applied to the recording system.

b) The initial E-plane of the test signal radiated by the test transmitter through its antenna shall be horizontal. The EUT antenna main beam axis shall be aligned with the main beam axis of the test transmitter. The EUT antenna or its polarizer shall be rotated around its main beam axis such that its E-plane coincides with the E-plane of the test transmitter.

c) After any change of the E-plane of the test signal, the EUT antenna shall be rotated around its main beam axis such that its E-plane coincides with the E-plane of the test transmitter.

d) The frequency of the test signal shall be set 5 MHz above the bottom of the lowest frequency band as defined for the intended use of the MES.

e) The EUT shall be aligned to maximize the received signal and the recording system adjusted to give the maximum reading on the chart.

f) The EUT shall be driven in azimuth through 180°.

g) The transmit pattern measurement is then obtained by driving the EUT in azimuth through 360° with the plotter recording the results.

h) The tests in c) to g) shall be repeated with the frequency changed to the middle of the lowest frequency band as defined for the intended use of the MES.

i) The tests in c) to g) shall be repeated with the frequency changed to 5 MHz below the top of the lowest frequency band as defined for the intended use of the MES.

j) The tests in c) to i) may be performed simultaneously.

k) The tests in c) to j) shall be repeated with the E-plane being vertical.

l) The tests in c) to j) shall be repeated with the E-plane at $+\alpha$ to the vertical plane. $\alpha°$ is defined as the worst case angle between the horizontal plane and the geostationary orbital arc, as seen from the range of latitudes as defined for the intended use of the MES.

m) The tests in c) to j) shall be repeated with the E-plane at $-\alpha°$ to the horizontal. $\alpha°$ is as defined in m).

n) The tests in c) to m) shall be repeated for all frequency bands as defined for the intended use of the MES.

6.3.3.4 Computation of results

The results shall be computed by producing a "mask" to the specified limits with the reference level being equal to the sum of the transmitter output power density, the gain of the antenna and twice the static rms pointing accuracy. This reference shall then be placed on the maximum point of the plot obtained from the transmit radiation pattern measurement, so as to ascertain that the off-axis EIRP density is within the mask, and thus conforming to the specification.

6.3.4 Measurement of the off-axis EIRP with the antenna

6.3.4.1 General

The requirements set out in clauses 6.1.1 to 6.1.3.1.2 for the measurements above 1 000 MHz shall apply.

6.3.4.2 Maximum EIRP density per 40 kHz ratio relative to the EIRP

6.3.4.2.0 General

For purpose of the test, the EUT shall be the MES fitted with its antenna.

6.3.4.2.1 Method of measurement

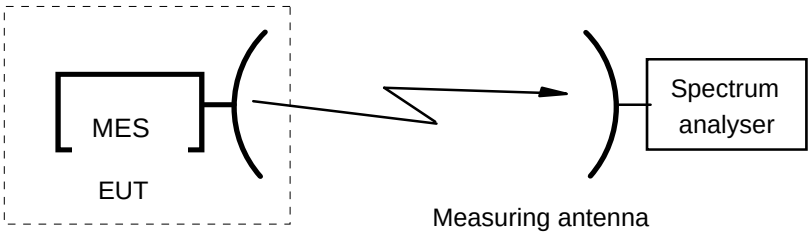


Figure 14: Test arrangement - radiated power density measurement

a) The test arrangement shall be as shown in figure 14, both antennas being in line of sight of each other.

b) The carrier shall be modulated by a pseudorandom bit sequence. The transmission shall be continuous where possible.

c) The resolution bandwidth of the spectrum analyser shall be set larger but as close as possible to the occupied bandwidth of the transmitted signal. The total power P1 received shall be measured in dBW.

d) The resolution bandwidth of the spectrum analyser shall be set as close as possible to the specified measuring bandwidth: 40 kHz. If the resolution bandwidth is different from the specified bandwidth then bandwidth linear correction shall be performed. The maximum value P2 of the power received in any 40 kHz bandwidth over the occupied bandwidth shall be measured in dBW.

e) The maximum EIRP density per 40 kHz ratio relative to the EIRP, in dBc/40 kHz, is the difference (P1 - P2).

6.3.4.3 Maximum on-axis EIRP

6.3.4.3.1 General

For purpose of the test, the EUT is the MES fitted with its antenna.

The distance between the EUT or the substitution antenna and the measuring antenna shall be such that the radiating near-field of each antenna shall not overlap with that of the other. The larger radiating near-field of the EUT and substitution antenna shall be used to determine the minimum distance between the EUT and measuring antenna in the first instance.

6.3.4.3.2 Test site

This test shall be performed using the method of measurement and test arrangement specified in the following clauses.

The tests can be carried out in an Open Area Test Site (OATS) environment or in an anechoic chamber or in a semi-anechoic chamber as they lead to comparable test results for establishing compliance. Ambient noise levels shall be at least 6 dB below the applicable unwanted emissions limit. The preferred test site is the semi-anechoic chamber.

For both the open area test site and the semi-anechoic chamber a metal ground plane shall be inserted on the natural ground plane and it shall extend at least 1 m beyond the perimeter of the EUT at one end and at least 1 m beyond the measurement antenna at the other end.

6.3.4.3.3 Method of measurement

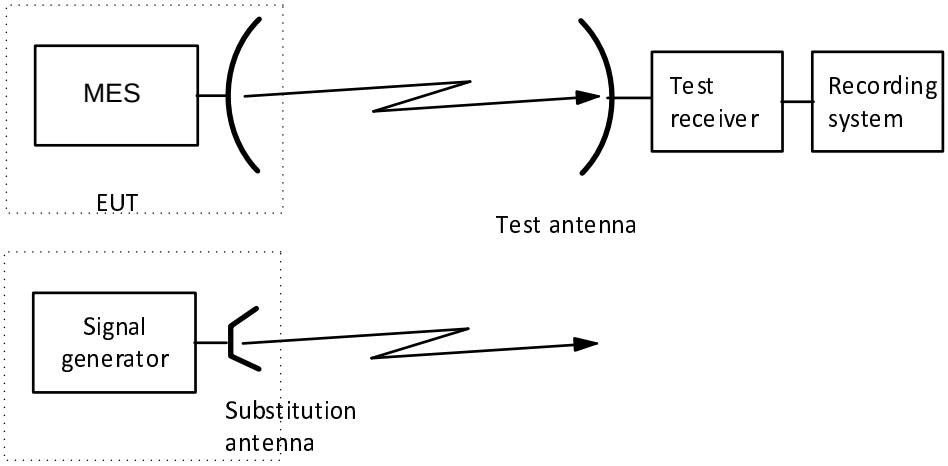


Figure 15: Test arrangement - maximum on-axis EIRP received measurement

a) The test arrangement shall be as shown in figure 15, both antennas being in line of sight of each other. A signal proportional to the angular position from the servo mechanism and the signal level from the test receiver proportional to the power received in dBW shall be applied to recording system.

b) The carrier transmitted by the EUT shall be, at maximum power, and modulated by a pseudorandom bit sequence. The transmission shall be continuous where possible.

c) The resolution bandwidth of the spectrum analyser shall be set larger but as close as possible to the occupied bandwidth of the transmitted signal.

d) The initial E-plane of the test signal radiated by the EUT through its antenna shall be horizontal. The test receiver antenna main beam axis shall be aligned with the main beam axis of the EUT antenna. The test receiver antenna or its polarizer shall be rotated around its main beam axis such that its E-plane coincides with the E-plane of the EUT.

e) After any change of the E-plane of the test signal, the test receiver antenna shall be rotated around its main beam axis such that its E-plane coincides with the E-plane of the EUT.

f) The frequency of the test signal shall be set 5 MHz above the bottom of the lowest frequency band as defined for the intended use of the MES.

g) The EUT shall be aligned to maximize the received signal and the recording system adjusted to give the maximum reading on the chart.

h) The EUT shall be driven in azimuth in one direction through 10°.

i) The pattern measurement is then obtained by driving the EUT in azimuth back through boresight to 10° the other side with the plotter recording the results.

j) The EUT shall be replaced by the signal generator connected to a calibrated antenna (a substitution antenna), transmitting a carrier at a frequency equal to the EUT carrier frequency. The received signal level shall be maximized.

k) This level shall be recorded on the recording system.

l) The signal generator with the substitution antenna shall be driven in azimuth as in h) and i).

m) The EIRP of the signal radiated by the EUT shall be calculated from:

$$EIRP_{EUT} = L_1 - L_2 + G + P$$

where:

- $EIRP_{EUT}$ is the EIRP of the signal radiated by the EUT (dBW), in the considered direction;

- L_1 is the level obtained with the EUT (dBW), in the considered same direction;
- L_2 is the level obtained with the signal generator with the substitution antenna (dBW);
- G is the calibrated gain of the substitution antenna at the test frequency (dBi);
- P is the power produced by the signal generator at the flange of the substitution antenna (dBW).

n) The tests in g) to m) shall be repeated with the frequency changed to the middle of the lowest frequency band as defined for the intended use of the MES.

o) The tests in g) to m) shall be repeated with the frequency changed to 5 MHz below the top of the lowest frequency band as defined for the intended use of the MES.

p) The tests in g) to o) may be performed simultaneously.

q) The tests in g) to p) shall be repeated with the E-plane vertical.

r) The tests in g) to p) shall be repeated with the E-plane at +45° to the horizontal plane.

s) The tests in g) to p) shall be repeated with the E-plane at -45° to the horizontal plane.

t) The tests in g) to s) shall be repeated for all frequency bands as defined for the intended use of the MES.

u) The maximum on-axis EIRP of the signal radiated by the EUT is the maximum value of the values calculated in step m).

6.3.4.4 Antenna transmit radiation pattern

6.3.4.4.1 General

For the purpose of the present document, the antenna transmit radiation pattern is a diagram relating field strength to direction relative to the pointing angle of the antenna at a constant large distance from the antenna (in the far field of the antenna).

For purpose of the test, the EUT shall be the MES fitted with its antenna.

The distance between the EUT or the substitution antenna and the measuring antenna shall be such that the radiating near-field of each antenna shall not overlap with that of the other. The larger radiating near-field of the EUT and substitution antenna shall be used to determine the minimum distance between the EUT and measuring antenna in the first instance.

6.3.4.4.2 Test site

This test shall be performed using the method of measurement and test arrangement specified in the following clauses.

The tests can be carried out in an Open Area Test Site (OATS) environment or in an anechoic chamber or in a semi-anechoic chamber as they lead to comparable test results for establishing compliance. Ambient noise levels shall be at least 6 dB below the applicable unwanted emissions limit. The preferred test site is the semi-anechoic chamber.

For both the open area test site and the semi-anechoic chamber a metal ground plane shall be inserted on the natural ground plane and it shall extend at least 1 m beyond the perimeter of the EUT at one end and at least 1 m beyond the measurement antenna at the other end.

6.3.4.4.3 Method of measurement

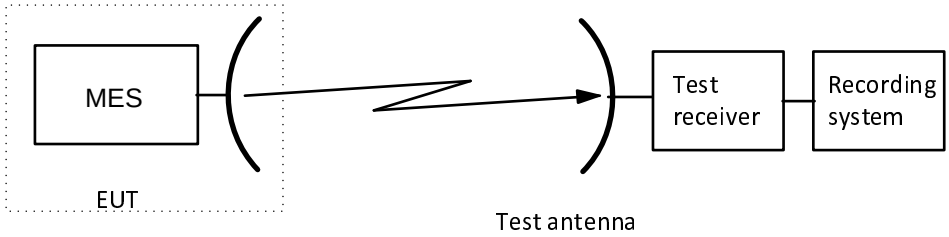


Figure 16: Test arrangement - antenna transmit radiation pattern measurement

a) The test arrangement shall be as shown in figure 16, both antennas being in line of sight of each other. A signal proportional to the angular position from the servo mechanism and the signal level from the test receiver proportional to the power received in dBW shall be applied to the recording system.

b) The carrier transmitted by the EUT shall be modulated by a pseudorandom bit sequence. The transmission shall be continuous where possible.

c) The resolution bandwidth of the spectrum analyser shall be set larger but as close as possible to the occupied bandwidth of the transmitted signal.

d) The initial E-plane of the test signal radiated by the EUT through its antenna shall be horizontal. The test receiver antenna main beam axis shall be aligned with the main beam axis of the EUT. The test receiver antenna or its polarizer shall be rotated around its main beam axis such that its E-plane is coincides with the E-plane of the EUT.

e) After any change of the E-plane of the EUT test signal, the test receiver antenna shall be rotated around its main beam axis such that its E-plane is coincides with the E-plane of the EUT.

f) The frequency of the EUT test signal shall be set 5 MHz above the bottom of the lowest frequency band as defined for the intended use of the MES.

g) The EUT shall be aligned to maximize the received signal and the recording system adjusted to give the maximum reading on the chart.

h) The EUT shall be driven in azimuth through 180°.

i) The transmit pattern measurement is then obtained by driving the EUT in azimuth through 360° with the plotter recording the results.

j) The tests in g) to i) shall be repeated with the frequency changed to the middle of the lowest frequency band as defined for the intended use of the MES.

k) The tests in g) to i) shall be repeated with the frequency changed to 5 MHz below the top of the lowest frequency band as defined for the intended use of the MES.

l) The tests in g) to k) may be performed simultaneously.

m) The tests in e) to l) shall be repeated with the E-plane horizontal.

n) The tests in e) to l) shall be repeated with the E-plane at $+\alpha°$ to the horizontal plane. $\alpha°$ is defined as the worst case angle between the horizontal plane and the geostationary orbital arc, as seen from the range of latitudes as defined for the intended use of the MES.

o) The tests in e) to l) shall be repeated with the E-plane at $-\alpha°$ to the horizontal plane. $\alpha°$ is as defined in n).

p) The tests in e) to o) shall be repeated for all frequency bands as defined for the intended use of the MES.

6.3.4.5 Computation of results

The results shall be computed by producing a "mask" to the specified limits with the reference level being equal to the sum of the transmitter maximum on-axis EIRP, the maximum EIRP density per 40 kHz ratio to the EIRP, and twice the static rms pointing accuracy. This reference shall then be placed on the maximum point of the plot obtained from the transmit radiation pattern measurement, so as to ascertain that the off-axis EIRP density is within the mask, and thus conforming to the specification.

6.4 Control and monitoring

6.4.0 General

If the EUT is an MES that has been modified consistent with its intended use for these tests then full documentation of such modification(s) shall be provided to prove that the modification(s) will simulate the required test condition.

For the purpose of this test the EUT is defined as the MES either with or without its antenna connected.

The measurement of the EIRP spectral density shall be limited to within either the nominated bandwidth or a 10 MHz bandwidth centred on the carrier frequency, whichever is the greater.

6.4.1 Test arrangement

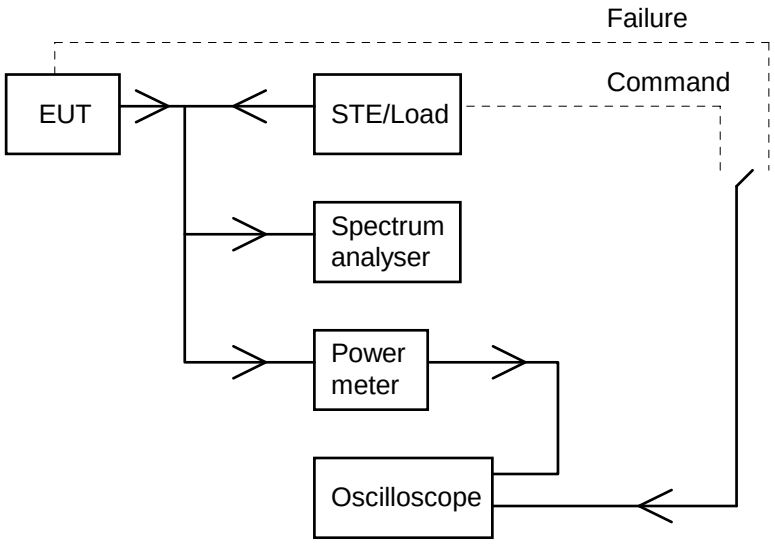


Figure 17: General test arrangement for control and monitoring tests for conducted measurements

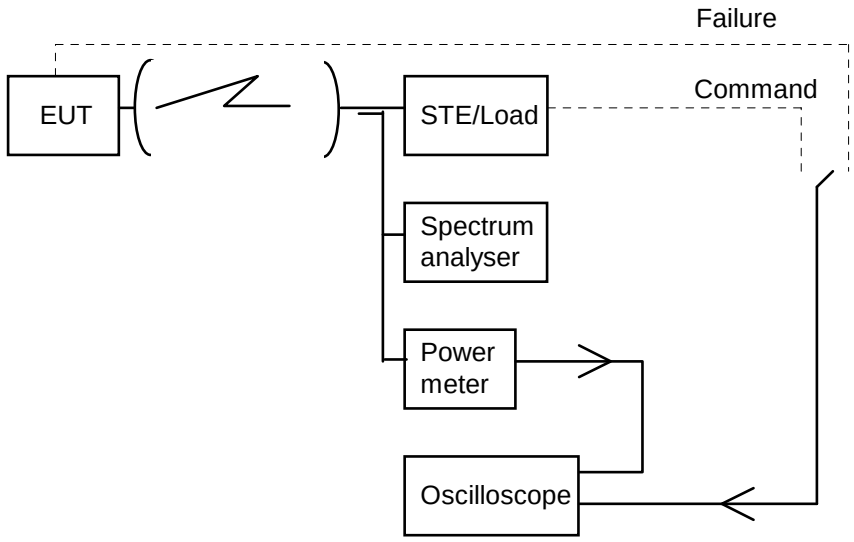


Figure 18: General test arrangement for control and monitoring tests for radiated measurements

The test arrangement shall be as shown in figure 17 or 18. The EUT shall be authorized to transmit and shall be in the carrier-on state at the commencement of each test. The dual trace storage oscilloscope shall monitor by measuring the time difference between the command, or failure, and the occurrence of the expected event (e.g. the transmission suppression). The power meter and spectrum analyser shall monitor the EUT output level.

6.4.2 Processor monitoring

6.4.2.1 Test method

a) Each of the processors within the EUT shall, in turn, be caused to fail.

b) Within 1 s of such failure the EUT shall cease to transmit as measured by the oscilloscope.

c) The power meter and spectrum analyser shall be observed to ascertain that the transmissions have been suppressed.

d) The failed processor shall be restored to normal working condition and the EUT shall be restored to normal working before the next processor shall be induced to fail.

6.4.3 Transmit subsystem monitoring

6.4.3.1 Test method

a) The frequency generation sub-system shall be caused to fail in respect of:

   1) Frequency stability.

   2) Output.

b) Within 6 s of such failure the EUT shall cease to transmit as measured by the oscilloscope.

c) The power meter and spectrum analyser shall be observed to ascertain that the transmissions have been suppressed.

d) The frequency generation sub-system shall be restored to normal working condition and the EUT shall be restored to normal working before the next induced failure.

6.4.4 Power-on/Reset

6.4.4.1 Test method

a) The EUT shall be switched off and the STE shall not transmit the CC.

b) The EUT shall be switched on.

c) The EUT shall not transmit during and after switching-on, and shall enter the carrier-off state.

The events from a) to c) shall be displayed and verified with the oscilloscope and by measurement of the transmitted signal. If a manual reset function is implemented the following test shall be performed:

d) The EUT shall be switched on and the STE shall transmit the CC.

e) A call shall be initiated from the EUT and the EUT shall enter the carrier-on state.

f) The reset function shall be initiated.

g) The EUT shall enter the carrier-off state.

The event from e) to g) shall be displayed and verified with the oscilloscope and by measurement of the transmitted signal.

6.4.5 Control Channel (CC) reception

6.4.5.1 Test method

The following tests shall be performed:

- case where the CC has never been received by the EUT;
- case where the CC is lost by the EUT during a transmission period of a call;
- case where the CC is lost by the EUT during a period without transmission;
- case where the CC is being lost by the EUT and a call is initiated within the Time-Out period T1.

The Time-Out period T1 used in the tests shall be 30 seconds.

a) Case where the CC has never been received by the EUT:

   a1) The EUT shall be switched off and the STE shall not transmit the CC.

   a2) The EUT shall be switched on.

   a3) A call shall be initiated from the EUT.

   a4) The EUT shall remain in the carrier-off state.

The events from a2) to a4) shall be displayed and verified with the oscilloscope and by measurement of the transmit signal.

b) Case where the CC is lost by the EUT during a transmission period of a call:

   b1) The EUT shall be switched-on and the STE shall transmit the CC.

   b2) A call shall be initiated from the EUT.

   b3) The STE shall stop transmitting the CC.

   b4) Within T1 from b3), the EUT shall enter the carrier-off state.

The events from b2) to b4) shall be displayed and verified with the oscilloscope and by measurement of the transmitted signal.

c) Case where the CC is lost by the EUT during a period without transmission:

c1) The EUT shall be switched on and the STE shall transmit the CC.

c2) The STE shall stop transmitting the CC.

c3) More than T1 later, a call shall be initiated from the EUT.

c4) The EUT shall remain in the carrier-off state.

The events from c2) to c4) shall be displayed and verified with the oscilloscope and by measurement of the transmitted signal.

d) Case where the CC is being lost by the EUT and a call is initiated within the T1 period:

d1) The EUT shall be switched on and the STE shall transmit the CC.

d2) The STE shall stop transmitting the CC.

d3) Within the period T1 from d2), a call shall be initiated from the EUT.

d4) The EUT may transmit but within the T1 period the EUT shall enter the carrier-off state.

The events from d2) to d4) shall be displayed and verified with the oscilloscope and by measurement of the transmitted signal.

6.4.6 Network control commands

6.4.6.1 Test method

The following tests shall be performed in sequence:

- Transmission enable command.
- Transmission disable command.
- Identification request.

a) Transmission enable command:

a1) The EUT shall be switched-on and the STE shall transmit the CC.

a2) The EUT shall enter the carrier-off state.

a3) A call shall be initiated from the EUT, the EUT shall remain in the carrier-off state.

a4) The STE shall transmit an enable command to the EUT.

a5) A call shall be initiated from the EUT.

a6) The EUT shall enter the carrier-on state and shall transmit.

The events from a2) to a6) shall be displayed and verified with the oscilloscope and by measurement of the transmitted signal.

b) Transmission disable command:

b1) Continue from a6).

b2) The STE shall transmit a disable command to the EUT.

b3) The EUT shall enter the carrier-off state within 1 s.

b4) A call shall be initiated from the EUT.

b5) The EUT shall remain in the carrier-off state.

b6) The STE shall transmit an enable command.

b7) A call shall be initiated from the EUT.

b8) The EUT shall enter the carrier-on state and shall transmit.

b9) The call shall be terminated at the EUT.

The events from b2) to b9) shall be displayed and verified with the oscilloscope and by measurement of the transmitted signal.

c) Identification request:

c1) Continue from b9).

c2) The STE shall transmit an identification request.

c3) The EUT shall enter the carrier-on state and shall transmit its identification code.

The STE shall display the identification code sent by the EUT.

6.4.7 Initial burst transmission

6.4.7.1 Test method

a) The EUT shall be switched-off and the STE shall transmit the CC.

b) The EUT shall be switched-on.

c) The EUT shall not transmit, except the initial bursts, if any.

d) Each initial burst shall not last more than 1 second, and the transmission of the initial bursts shall not exceed 1 % of the time.

The events from b) to d) shall be displayed and verified with the oscilloscope and by measurement of the transmitted signal.

6.5 Receive antenna off-axis gain pattern

6.5.1 Test method

6.5.1.1 Test site

This test shall be performed using the method of measurement and test arrangement specified in the following clauses.

The tests can be carried out in an Open Area Test Site (OATS) environment or in an anechoic chamber or in a semi-anechoic chamber as they lead to comparable test results for establishing compliance. Ambient noise levels shall be at least 6 dB below the applicable unwanted emissions limit. The preferred test site is the semi-anechoic chamber.

6.5.1.2 Method of measurement

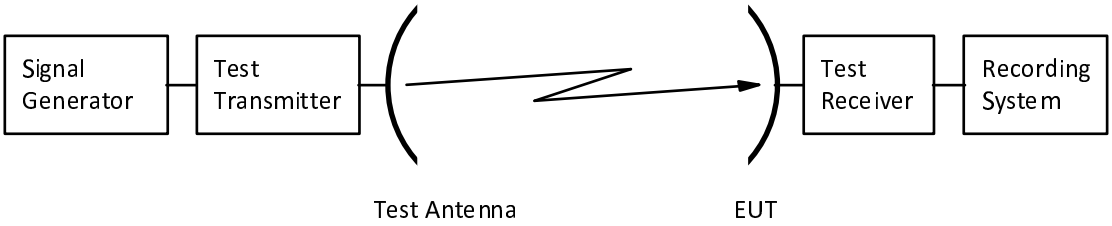


Figure 19: Test arrangement - antenna receive pattern measurement

a) The test arrangement shall be as shown in figure 19 with the EUT connected to the test receiver.

b) A signal proportional to the angular position from the servo mechanism and the signal level from the test receiver shall be applied to the recording system.

c) The test frequencies shall be the centre frequency of each applicable frequency range. The E-plane shall be vertical.

d) The EUT shall be aligned to maximize the received signal level and the recording system shall be adjusted to give the maximum reading on the chart.

e) The EUT shall be driven in azimuth through 180°.

f) The pattern measurement is then obtained by driving the EUT in azimuth through 360° with the plotter recording the results.

g) The tests in b) to e) shall be repeated with the frequency changed to the lower limit of the applicable band as defined for the intended use of the MES.

h) The tests in b) to e) shall be repeated with the frequency changed to the upper limit of the applicable band as defined for the intended use of the MES.

i) The tests in b) to h) shall be repeated with the frequencies changed to the others specified if the design of the equipment is such that operation is possible, but not necessarily simultaneously, in all bands.

j) The tests in b) to h) shall be repeated with the test signal being transmitted in the H-plane instead of the E-plane.

k) The tests in b) to h) shall be repeated with the test signal being transmitted in a plane at 45° to the H-plane.

l) The tests in b) to h) shall be repeated with the test signal being transmitted in a plane at 90° to that in k).

m) The tests in b) to l) shall be repeated between the angles of φ_r and 7° with the EUT rotated through 90°, or the test antenna or the polarization subsystem of the EUT rotated by 90°, to give the cross-polar measurement.

6.5.1.3 Computation

The computation of results shall be carried out by producing a "mask" to the specified limits with the reference level being equal to the gain of the antenna. This reference shall then be placed on the maximum point of the plot obtained from the pattern measurements.

6.6 Blocking performance

6.6.1 Test method

a) The output signals of two signal generators shall be combined with equal weight. The first signal shall be the wanted modulated signal as defined for the intended use, tuned at the centre frequency of the receive frequency band and have a level in the operational range. The second signal shall be a CW signal with same level. The combined signal shall be coupled to the LNB input A spectrum analyser shall be connected to the LNB output in a way that allows to supply the LNB with power.

b) f_c is the center frequency of the receive frequency band.

c) The first signal generator frequency shall be set to f_c.

d) The first signal generator level shall be set to a level in the LNB operational input level range.

e) The spectrum analyser shall be set for measuring the level of the converted first signal at the LNB output.

f) The second signal generator frequency shall be set to $f_c - 20$ MHz or any other frequency outside the occupied bandwidth

g) The second signal generator level shall be adjusted so that the measured level is 1 dB less than in absence of the second signal.

h) The second signal generator level shall be noted down as reference level.

i) The second signal generator frequency shall be set to the frequency of interest.

j) The second signal generator level shall be adjusted so that the measured level is 1 dB less than in absence of the second signal.

k) The rejection at the frequency of interest is equal to the second signal generator level minus the reference level determined in step i).

l) Steps j) to l) shall be repeated for frequencies in the ranges of table 3.

NOTE: The worst case rejection in a particular frequency range can be determined after step i) by sweeping the second signal generator frequency over the frequency range and observe the gain compression, then perform steps j) to l) with the frequency where gain compression is highest.

6.7 Adjacent Signal Selectivity

6.7.1 Test method

a) Two test signal generators shall be used, each generating the same modulated signal as the wanted signal in the IME input frequency range and thermal noise.

b) The signal generators shall be connected to the IME input through a splitter (combiner).

c) The test signal generators shall be set to the frequencies and levels according to table 4.

d) The IME shall be set to receive the signal of the first test signal generator.

e) The second test signal generator shall be set to signal off.

f) The noise level (or signal to noise ratio) of the first test signal generator shall be varied in order to determine the threshold reference sensitivity level.

g) The second signal generator shall be set to signal on.

h) The noise level (or signal to noise ratio) of the first test signal generator shall be varied in order to determine the threshold reference sensitivity level.

i) The degradation is equal to the noise level (or signal to noise ratio) determined in step h minus that determined in step f.

j) The result is the highest degradation found.

Annex A (informative): Relationship between the present document and the essential requirements of Directive 2014/53/EU

The present document has been prepared under the Commission's standardisation request C(2015) 5376 final [i.3] to provide one voluntary means of conforming to the essential requirements of Directive 2014/53/EU on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC [i.6].

Once the present document is cited in the Official Journal of the European Union under that Directive, compliance with the normative clauses of the present document given in table A.1 confers, within the limits of the scope of the present document, a presumption of conformity with the corresponding essential requirements of that Directive, and associated EFTA regulations.

Table A.1: Relationship between the present document and the essential requirements of Directive 2014/53/EU

Harmonised Standard ETSI EN 301 427					
Requirement				Requirement Conditionality	
No	Description	Essential requirements of Directive	Clause(s) of the present document	U/C	Condition
1	Unwanted emissions outside the 14,00 GHz to 14,25 GHz band	3.2	4.2.1	U	
2	Unwanted emissions within the 14,00 GHz to 14,25 GHz band	3.2	4.2.2	U	
3	Off-axis EIRP density in the Nominated bandwidth	3.2	4.2.3	U	
4	Processor monitoring	3.2	4.2.4.1	U	
5	Transmit subsystem monitoring	3.2	4.2.4.2	U	
6	Power-on/Reset	3.2	4.2.4.3	U	
7	Control Channel (CC) reception	3.2	4.2.4.4	U	
8	Network control commands	3.2	4.2.4.5	U	
9	Initial burst transmission	3.2	4.2.4.6	U	
10	Receive Antenna off-axis gain pattern	3.2	4.2.5	C	Satellite communications networks may contain a central control unit that keeps received interference low by taking into account the actual antenna gain pattern of earth stations. The requirement is not relevant for earth stations that are always operated as part of such a network. The requirement is relevant in all other cases. If the receive antenna performance does not meet the requirement of clause 4.2.5, then the earth station accepts the risk of additional co-frequency interference caused by its off-axis gain that is not compliant with the mask.
11	Receiver blocking. Blocking performance	3.2	4.2.6	U	
12	Receiver selectivity. Adjacent Signal Selectivity	3.2	4.2.7	U	

Harmonised Standard ETSI EN 301 427					
Requirement				Requirement Conditionality	
No	Description	Essential requirements of Directive	Clause(s) of the present document	U/C	Condition
13	Receiver sensitivity	3.2			There is no causal relationship between receiver sensitivity and interference in the case of satellite communications. Therefore, a quantitative calculation is not possible. Requirements are taken into account in the satellite link budget for the intended use.
14	Receiver co-channel rejection	3.2			The impact of an unwanted signal is taken into account in the link budget apportionment to interference degradation of the link budget margin.
15	Receiver spurious response rejection	3.2			Frequencies other than the image frequency are not relevant because this is covered by receiver blocking and adjacent channel requirements which are already included in the receiver parameters.
16	Receiver radio-frequency intermodulation	3.2			In an FSS network, receiver radio-frequency intermodulation is not relevant, because possibly inter-modulating signals originating from other networks are rejected thanks to requirements on receive antenna off-axis gain pattern and adjacent signal selectivity. Interfering signals up to 28 dB above the wanted carrier can be tolerated with minimal impact on performance. Concerning signals originating from the own network, the network operator has control over levels and can prevent intermodulation.
17	Receiver dynamic range	3.2			In an FSS network, receiver dynamic range has no influence on harmful interference. The satellite earth stations are deployed in a manner that results in a situation where it is impossible to receive a wanted signal that is high enough to produce any overloading effect (satellite systems are power limited). So that it is not necessary to specify receiver dynamic range. See clause 5.3.6.1 of ETSI EG 203 336 [i.4].

Harmonised Standard ETSI EN 301 427					
Requirement				Requirement Conditionality	
No	Description	Essential requirements of Directive	Clause(s) of the present document	U/C	Condition
18	Reciprocal mixing	3.2			For FSS equipment, the parameter is excluded for the following reason. It is considered that the reciprocal mixing effects are implicitly covered in HSs where comprehensive interference characteristics are specified in terms of selectivity and/or blocking requirements, thus removing the need for this parameter to be included in HSs as the effects of receiver selectivity and reciprocal mixing cannot be separated. See clause 5.3.6.2 of ETSI EG 203 336 [i.4].
19	Desensitization	3.2			As desensitization is a receiver effect addressed by other parameters (including receiver blocking), its inclusion as a separate parameter in an HEN is not required. See clause 5.3.2.3 of ETSI EG 203 336 [i.4].

Key to columns:

Requirement:

No A unique identifier for one row of the table which may be used to identify a requirement.

Description A textual reference to the requirement.

Essential requirements of Directive

Identification of article(s) defining the requirement in the Directive.

Clause(s) of the present document

Identification of clause(s) defining the requirement in the present document unless another document is referenced explicitly.

Requirement Conditionality:

U/C Indicates whether the requirement is unconditionally applicable (U) or is conditional upon the manufacturer's claimed functionality of the equipment (C).

Condition Explains the conditions when the requirement is or is not applicable for a requirement which is classified "conditional".

Presumption of conformity stays valid only as long as a reference to the present document is maintained in the list published in the Official Journal of the European Union. Users of the present document should consult frequently the latest list published in the Official Journal of the European Union.

Other Union legislation may be applicable to the product(s) falling within the scope of the present document.

Annex B (informative): Maximum measurement uncertainty

The measurements described in the present document are based on the following assumptions:

- the measured value related to the corresponding limit is used to decide whether an equipment meets the requirements of the present document;
- the value of the measurement uncertainty for the measurement of each parameter is included in the test report.

Table B.1 shows the recommended values for the maximum measurement uncertainty figures (on the basis of ETSI TR 100 028-1 [i.5]).

Table B.1: Maximum measurement uncertainties

Parameter	Uncertainty
Radio frequency	±10 kHz
RF power	±0,75 dB
Conducted Measurements for spurious	±4 dB
Radiated measurements for spurious	±6 dB
Antenna gain	±2 dB

Annex C (informative): Bibliography

- ETSI ETS 300 255 (1994): "Satellite Earth Stations and Systems (SES); Land Mobile Earth Stations (LMESs) operating in the 11/12/14 GHz bands providing Low Bit Rate Data Communications (LBRDC)".
- ETSI ETS 300 456 (1995): "Satellite Earth Stations and Systems (SES); Test methods for Very Small Aperture Terminals (VSATs) operating in the 11/12/14 GHz frequency bands".
- EN 55022 (1994): "Limits and methods of measurement of radio disturbance characteristics of information technology equipment", (produced by CEN).
- ETSI SR 001 470: "Guidance to the production of candidate Harmonized Standards for application under the R&TTE Directive (1999/5/EC); Pro-forma candidate Harmonized Standard".

Annex D (informative): Change history

Version	Information about changes
V2.1.2	Contribution for review by SES HARM#80 to address requirements of RED
V2.1.3	Modifications introduced by SES HARM#80
V2.1.4	Contribution to SES HARM #81 adding Annex A and overall editorial check
V2.1.5	Modifications at SES HARM#81 on non-directional antennas and preferred methods for test sites
V2.1.6	Editorial cleaning, removal highlighted paragraphs by Rapporteur after SES HARM#81
V2.1.7	Further editorial update of the acronym "MES" reinserting in the text
V2.1.8	Modifications of the language " intended use" by "intended use of the MES"
V2.1.9	Review by SES HARM#84
V2.1.10	Review by SES HARM#88 (review description in the scope, update Annex A)
V2.1.11	Clean version after SES HARM88
V2.1.12	Further review addressing comments from ETSI secretariat
V2.1.13	Further review addressing comments from ETSI secretariat and alignment text with other ENs after WG HARM 99
V2.1.14 to V2.1.19	Further revisions by WG HARM to align texts with other ENs, in particular on references to test sites. Latest version as result of WG HARM 121
V2.1.21	Final version reviewed by WG HARM 123
V2.1.22	Post edited version on formatting prepared for TC SES 109 approval
V2.1.23	Review by WG HARM 147 addressing reference to preferred test site.
V2.1.24	Review by WG HARM 148 updating reference to Open Area Test Site (OATS)

History

Version	Date	Status
V1.1.1	May 2000	Publication
V1.2.1	November 2001	Publication
V2.1.1	June 2016	Publication
V2.1.25	September 2026	SRdAP process EV 20261214: 2026-09-21 to 2026-12-14